A Multi-Stream Temporal Context Model for Narrative Memory

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Abstract

The Temporal Context Model[10] and its extension the Context Maintenance and Retrieval model[15] together provide a widely-used computational framework for understanding how memory for a sequence of events depends on the temporal context in which they were encoded. In both, a single temporal-context vector drifts continuously through encoding, so the similarity between the contexts of two events falls off smoothly with the number of intervening events. This single-stream assumption explains a broad range of phenomena in word-list free recall, but it faces a puzzle when memory for *naturalistic* narratives is studied. In pilot data from a cued-recall study of a television show whose plot interleaves three storylines, Xu et al.[20] find that cued recall of two narratively adjacent events is essentially unchanged by whether they are temporally adjacent (the *grouped* condition) or are separated by 2 to 7 intervening events from other storylines (the *bridge* condition), with a Bayes factor of $\mathrm{BF}_{01} = 412$ for the null of no difference. Under a single-stream context model, the bridge condition should incur substantial context drift and therefore a substantial cued-recall deficit – the observed equivalence is unexpected. Here we propose a *Multi-Stream Temporal Context Model* (MS-TCM) that augments the single global temporal context with a bank of storyline-specific context vectors that drift only while their own storyline is active and are held constant ("frozen") otherwise. We derive closed-form parameter regimes under which MS-TCM predicts the grouped–bridge equivalence, and fit the model by

maximum likelihood, with participant-level bootstrap confidence intervals, to the category condition of the feature-rich free-recall dataset of Manning et al.[13] as a worked example. The MLE strongly prefers MS-TCM over a single-stream baseline (ΔAIC = 253) and lands in the parameter regime predicted by the whitepaper: a dominant storyline-context weight with negligible global weight. We discuss what this fit does and does not tell us about the model's adequacy, and outline five testable predictions that MS-TCM generates for naturalistic memory experiments beyond the free-recall paradigm.

1 The puzzle of memory across narrative storylines

Recency and contiguity in free recall. Two robust principles organize human memory for a sequence of events. The first is *recency*: items encountered late in a sequence are remembered better than items encountered earlier. The second is *contiguity*: successful recall of one item biases the next recall toward items that were encoded close to it in time[10,12,14]. Both principles vary smoothly with the temporal separation between events. In free recall of word lists, contiguity is most clearly visible in the conditional response probability as a function of lag (lag-CRP), which peaks at lag +1 (the immediately-following item) and decays for larger lags[11]. The same principles show up in much longer time scales: in continuous-distractor free recall, a recency-like decay unfolds over tens of seconds[3,9].

The *Temporal Context Model* (TCM;[10]) offers a simple explanation for both principles. A single d-dimensional temporal context vector $\mathbf{c}(t)$ is maintained across encoding. Each encoded item f_i at time t updates the context as

$$\mathbf{c}(t) = \rho\, \mathbf{c}(t-1) + \beta\, \mathbf{c}_i^{\mathrm{IN}}, \tag{1}$$

where $\rho = \sqrt{1-\beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ under unit-norm inputs and $\beta \in (0, 1)$ is the drift rate. The context similarity $\mathbf{c}(t) \cdot \mathbf{c}(t')$ decays as $\rho^{|t-t'|}$, so items encoded near each other in time share more context than items encoded far apart; at retrieval, reinstatement of a cue's encoding context preferentially activates its temporal neighbours. TCM produces both recency (end-of-list items dominate retrieval because their context overlaps most with the retrieval cue) and contiguity (nearby items share context and are cross-cued). The *Context Maintenance and Retrieval* model (CMR;[15]) extends this machinery with a second sub-region of context that carries source or task information, and shows that CMR jointly captures semantic, temporal, and source clustering.

Memory for interleaved storylines. For decades the neighbourhood of phenomena around TCM and CMR was studied almost exclusively with experimenter-generated sequences of words. Over the last ten years, an increasingly rich literature has begun to ask how the same contextual principles apply when the sequence is a naturalistic narrative – a television show, a movie, or a spoken story – that humans watch or listen to in a continuous stream[1,8,22]. Two distinctive features of naturalistic narratives are relevant here.

First, *narratives are structured by events*. Event-segmentation theory[21] and the situation-models literature[23] argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, goals, or causal structure. Event boundaries modulate memory: items encoded across a boundary are remembered as further apart in time than items within the same event, even when objective time is matched[6,8]. Shared schema structure within an event supports robust reinstatement after interruption[2].

Second, *naturalistic narratives often interleave multiple storylines*. Many television shows and novels tell their stories by cutting between parallel threads: the viewer experiences, moment to moment, a single linear stream of events that is the braided-together interleaving of multiple ongoing storylines. *Why Women Kill* is one such show; *This Is Us* is another. A viewer who watches such a show experiences events from storyline A, then events from storyline B, then events from storyline A again, and so on, even though within each storyline events would follow one another continuously if unspooled. Figure 1 illustrates the metaphor: each storyline is a strand with its own internal timeline, and the viewer's experience is a braid that splices segments from the strands together.

The empirical puzzle. Xu et al.[20] asked whether cued recall of narrative events obeys the same temporal-contiguity principles that govern cued recall of word lists. Their participants watched the first season of *Why Women Kill* – a show that interleaves three storylines set in different historical periods – and were later cued with one event from the show and asked to recall the event that preceded or followed it. The cued-to-target pairs fell into three conditions:

1. **Within-event** (Experiment 2): cue and target belong to the same event (same storyline, same scene).
2. **Across-event-within-storyline** (Experiment 1, "grouped"): cue and target are consecutive events on the same storyline with no intervening events from other storylines.
3. **Across-event-bridge** (Experiment 3, "interleaved"): cue and target are narratively consecutive on the same storyline but separated by $m = 2$–7 intervening events from other storylines (mean ≈ 3).

A single-stream contiguity account predicts a substantial decrement for the bridge condition relative to the grouped condition: the bridge target is four times further away in the viewer's global temporal context than the grouped target, and under Equation 1 context similarity falls as ρ^{m+1}. In the data, by contrast, the bridge and grouped conditions are statistically indistinguishable, with a Bayes factor $\mathrm{BF}_{01} = 412$ in favour of the null hypothesis of no difference between them. This is the puzzle we seek to explain.

2 A multi-stream temporal context model

Parallel context streams. MS-TCM assumes that while a listener or viewer watches a narrative, the brain maintains *multiple* temporal-context vectors in parallel. One of these is a global context, $\mathbf{c}_G(t)$, that behaves exactly like TCM's single context: it drifts at every encoding step and therefore tracks the viewer's timeline (Figure 2, top centre; Figure 1C, top rail). The rest of the context bank is one vector per storyline, $\{\mathbf{c}_{Sk}(t)\}_{k=1}^{K}$, each of which drifts *only while its own storyline is the active storyline* and is held constant otherwise.

Concretely, when event i from active storyline S^* is encoded at global time step t:

$$\mathbf{c}_G(t) = \rho_G\,\mathbf{c}_G(t-1) + \beta_G\,\mathbf{c}_i^{\mathrm{IN}}, \tag{2}$$

$$\mathbf{c}_{S^*}(t) = \rho_S\,\mathbf{c}_{S^*}(t_{\mathrm{prev}}^{S^*}) + \beta_S\,\mathbf{c}_i^{\mathrm{IN}}, \tag{3}$$

$$\mathbf{c}_{Sk}(t) = \mathbf{c}_{Sk}(t-1) \quad \text{for all} \quad k \neq S^*, \tag{4}$$

where $\rho_G = \sqrt{1-\beta_G^2}$, $\rho_S = \sqrt{1-\beta_S^2}$, and $t_{\mathrm{prev}}^{S^*}$ is the most recent prior time step at which S^* was active. Equation 4 is the key architectural commitment of MS-TCM and the ingredient that standard TCM[10] and CMR[15] do not have: a storyline context's state at time t depends only on the events in its own storyline, not on intervening events from other storylines. When the storyline resumes at a later viewer time t', the update in Eq. 3 reuses the *frozen* state from $t_{\mathrm{prev}}^{S^*}$ rather than whatever the storyline context would have drifted to had it updated at every intervening step.

Encoding and retrieval composites. Each event i on storyline S^* is associated with a composite encoding context that mixes the global and active-storyline contexts:

$$\mathbf{c}_{\mathrm{comp}}(i) = w_G\,\mathbf{c}_G(t) + w_S\,\mathbf{c}_{S^*}(t), \qquad w_G + w_S = 1. \tag{5}$$

During cued recall, presentation of a cue event j on storyline S_j reinstates a retrieval composite:

$$\mathbf{c}_{\mathrm{ret}}(j) = w_G^{\mathrm{ret}}\,\mathbf{c}_G(t_j) + w_S^{\mathrm{ret}}\,\mathbf{c}_{Sj}(t_j), \qquad w_G^{\mathrm{ret}} + w_S^{\mathrm{ret}} = 1. \tag{6}$$

Crucially, the retrieval weights $(w_G^{\mathrm{ret}}, w_S^{\mathrm{ret}})$ need not equal the encoding weights (w_G, w_S). Task instructions can rebalance the cue: for instance, being asked to recall "what happened next in the same storyline" should plausibly increase w_S^{ret} relative to w_G^{ret}, amplifying the storyline context's contribution to the cue. The probability of recalling a candidate event i given cue j is a softmax-with-gain over a scored similarity,

$$s_i(j) = \cos\Big(\mathbf{c}_{\text{ret}}(j),\, \mathbf{c}_{\text{comp}}(i)\Big) \cdot a_i \cdot e^{-\lambda I_{ij}}, \qquad P(i \mid j) = \frac{\exp(\tau\, s_i(j))}{\sum_{k \in C(j)} \exp(\tau\, s_k(j))}, \tag{7}$$

where $a_i = \phi_s\, e^{-\phi_d(i-1)} + 1$ is the Polyn-et-al.-style primacy-gradient item bias, $\tau > 0$ is an inverse-temperature / softmax gain, I_{ij} is the cumulative between-event interference term scaled by λ (§4), and the candidate set $C(j)$ is the set of list items that have not yet been recalled on the current list (the standard TCM / CMR ”no repeats” convention). Both τ and (ϕ_s, ϕ_d) are fit alongside the other free parameters. Without τ the softmax of bounded cosine similarities in $[-1, 1]$ is close to uniform and the model fails to produce any of the shape features of free recall; without (ϕ_s, ϕ_d) there is no primacy mechanism and the serial- position curve has only a recency limb. Setting $\phi_s = 0$ recovers the vanilla formulation.

Relationship to TCM and CMR. MS-TCM specializes to standard TCM[10] when $w_S = 0$, in which case the storyline bank is inert and the model reduces to Eq. 1. The model is also superficially similar to CMR[15]: both augment TCM with a second source of contextual information beyond pure recency. But the architectures differ in two important ways.

First, CMR maintains a *single* context vector with multiple sub-regions (temporal and source), all of which update according to Eq. 1 at every step, with possibly different drift rates. MS-TCM maintains $K + 1$ *parallel context vectors* with qualitatively different update rules: only one of the K storyline contexts updates per step, and the other $K - 1$ are held constant. The frozen-while-inactive property is not a weighted-down version of drift; it is the absence of update.

Second, CMR's source sub-region is a *feature of the current event* – it represents the task or source under which the current item was encoded. MS-TCM's storyline contexts are *separate memory stores*: the storyline identity of an event determines which store gets updated, not what the store *contains*. A storyline context carries the accumulated contextual imprint of all previous events *on that storyline*, independent of whatever events have been encoded on other storylines in between.

3 Deriving the grouped–bridge equivalence

We now derive analytically the parameter regimes under which MS-TCM reproduces the observed grouped–bridge equivalence. Consider two consecutive storyline-S events i and i' encoded at viewer times t and $t' = t + m + 1$, so $m \geq 0$ intervening events from other storylines separate them. Under orthonormal item inputs (a standard simplification; see Eq. (1) of [10]), the per-stream context similarities after the second encoding are:

$$\cos(\mathbf{c}_G(t), \mathbf{c}_G(t')) \approx \rho_G^{m+1}, \tag{8}$$

$$\cos(\mathbf{c}_S(t), \mathbf{c}_S(t')) = \rho_S, \tag{9}$$

the first because $\mathbf{c}_G$ has drifted through $m + 1$ encoding steps, the second because $\mathbf{c}_S$ was frozen during the m intervening steps and so has undergone only one update between t and t'. A linear composite-similarity quantity $\sigma(m) \equiv w_G\, \rho_G^{m+1} + w_S\, \rho_S$ collapses the two conditions to

$$\sigma(0) = w_G\, \rho_G + w_S\, \rho_S \quad \text{(grouped)}, \qquad \sigma(m) = w_G\, \rho_G^{m+1} + w_S\, \rho_S \quad \text{(bridge)}. \tag{10}$$

The observed equivalence $\sigma(0) \approx \sigma(m)$ holds whenever one of three regimes obtains:

1. **Storyline-context dominance.** When $w_S \gg w_G$, both conditions are dominated by the $w_S\, \rho_S$ term, which is identical across conditions. The global drift contribution is a minority of the composite and the difference $w_G(\rho_G - \rho_G^{m+1})$ is accordingly small.

2. **Task-driven retrieval reweighting.** Even when encoding weights are comparable ($w_G \approx w_S$), the Experiment 3 instruction ("recall the next event in the same storyline") can plausibly upweight the storyline component at retrieval by increasing w_S^{ret} relative to w_G^{ret}, again shrinking the bridge–grouped gap.

3. **Matched interference.** The grouped condition pairs the target event with a within-storyline boundary event that shares characters and setting, which should produce substantial retroactive interference. The bridge condition pairs the target with events from *different* storylines,

which are categorically distinct and should produce less interference release from proactive interference;[18]. If this between-storyline material produces less interference than within-storyline material, the interference term $e^{-\lambda I_{ij}}$ in Eq. 7 can partially offset the reduced global context similarity in the bridge condition.

Numerical illustration. As a concrete anchor, at $\beta_G = \beta_S = 0.5$, $w_G = 0.2$, $w_S = 0.8$, and $m = 3$ (the mean number of intervening events in *Why Women Kill*), the closed-form composite similarities are $\sigma(0) = \sqrt{0.75} = 0.86603$ (grouped) and $\sigma(3) = 0.2 \cdot 0.5625 + 0.8\sqrt{0.75} = 0.80532$ (bridge), a difference of 0.06 that is small relative to typical item- and participant-level variance in cued-recall studies.[1]

4 Additional mechanisms

The core architecture in §2 is deliberately minimal – it is the smallest change to standard TCM that produces the frozen-while-inactive property required to explain the empirical puzzle. Three optional mechanisms can be added on top, each corresponding to a specific claim from the naturalistic-memory literature.

Resumption reinstatement (γ). When a storyline resumes after interruption, re-encountering its characters and setting may trigger automatic reinstatement of the storyline context to something close to its prior frozen value analogous to environmental-reinstatement effects in episodic memory;[7,17]. Formally, the storyline update becomes

$$\mathbf{c}_{S^*}(t) = \rho_S\, \mathbf{c}_{S^*}(t_{\text{prev}}^{S^*}) + \beta_S\, \mathbf{c}_i^{\text{IN}} + \gamma\, \mathbf{c}_{S^*}(t_{\text{prev}}^{S^*}), \tag{11}$$

where $\gamma \geq 0$ captures the reinstatement strength and reduces to Eq. 3 at $\gamma = 0$.

[1] The whitepaper for MS-TCM rounds $\sigma(3)$ to 0.806 via a rounding-cascade ($\rho \approx 0.866 \Rightarrow \rho^4 \approx 0.563 \Rightarrow \sigma(3) \approx 0.806$). The closed-form value is 0.80532..., which rounds to 0.805. The code's unit test `test_section_4_4_numerical_anchor` asserts the closed-form values at 10^{-6} tolerance. See Supplement §S2.

Conversational references (α_k). Xu et al.[19] show that characters in television narratives refer more often to past events on the same storyline than to future events. When storyline S resumes, explicit references in dialogue (*"Remember when we discussed the project?"*) can provide *explicit* reinstatement of earlier storyline events. These references enter the storyline update as additional inputs:

$$\mathbf{c}_{S^*}(t) = \rho_S \, \mathbf{c}_{S^*}(t_{\text{prev}}^{S^*}) + \beta_S \, \mathbf{c}_i^{\text{IN}} + \sum_k \alpha_k \, \mathbf{c}_{\text{ref}_k}^{\text{IN}}, \tag{12}$$

where ref_k indexes events explicitly referenced in the current event's dialogue and α_k controls the weight of each reference.

Differential interference (λ). The release-from-proactive-interference literature[18] suggests that intervening events from categorically distinct material produce less interference than within-category intervening material. MS-TCM captures this by scaling the retrieval score by $e^{-\lambda I_{ij}}$ (Eq. 7), where I_{ij} is the cumulative between-event interference from items encoded between i and j. For free recall we instantiate $I_{ij} = \max(0, |s_i - s_j| - 1)$, the count of items presented between candidate i (at serial position s_i) and cue j (at serial position s_j); $\lambda = 0$ disables the mechanism.

5 Testable predictions

Beyond the grouped–bridge equivalence, MS-TCM generates at least five predictions that discriminate it from single-stream accounts of narrative memory.

1. **Storyline-similarity manipulation.** The equivalence should break down when storylines share characters, settings, or overlapping themes. With highly similar storylines, distinct $\mathbf{c}_{Sk}$'s become harder to maintain as separable representations, reducing the effectiveness of the frozen-while-inactive mechanism. A natural test: compare bridge to grouped performance on *Why Women Kill* (three distinct storylines) against *The Affair* (two storylines showing the same events from two perspectives). MS-TCM predicts bridge ≈ grouped for the first; TCM predicts bridge < grouped in both.

2. **Number of intervening events (m).** Under MS-TCM, across-event-bridge performance should be relatively robust to m because storyline-context similarity is ρ_S regardless of m. A weak decline is still expected from the diminishing global-context contribution. In *This Is Us*, where m varies from 1 to 7, MS-TCM predicts a weak bridge-m slope and TCM predicts a steep one.

3. **Individual differences in working memory.** Maintaining K distinct storyline contexts in parallel should be more demanding than maintaining a single temporal context. Individuals with higher working-memory capacity should therefore show a stronger grouped $\approx$ bridge equivalence (and more sensitive $\mathbf{c}_{Sk}$ distinctions) than individuals with lower capacity.

4. **Temporal direction asymmetry.** Within-event cued recall in Xu et al.[20] shows a strong backward asymmetry ($\mathrm{BF}_{10} = 1.6 \times 10^{23}$) that the across-event conditions do not. MS-TCM attributes this to a reduction of pre-built backward links at event boundaries[6] Xu et al. 2024, since the storyline-context mechanism is approximately symmetric (ρ_S is the same regardless of direction). Manipulating pre-built links experimentally (e.g., by familiarizing participants with forward *or* backward event sequences before the cued-recall test) should dissociate the two mechanisms.

5. **First-recall latency.** Even with equivalent accuracy, bridge-condition responses may be *slower* than grouped- condition responses if retrieval begins with the (poorer) global cue and incorporates the storyline cue serially. A full-RT model of MS-TCM retrieval, along the lines of TCM-A[16] – not attempted here – would make this quantitative.

6 Worked example: fitting MS-TCM to a free-recall dataset

Because the cued-recall data from Xu et al.[20] are not yet publicly released, we demonstrate the MS-TCM machinery on a public free-recall dataset whose structure is analogous: the category condition of Manning et al.[13]'s feature-rich free-recall experiment. In this condition, $N = 30$ participants each study 16 lists of 16 words drawn from 4 taxonomic categories (Figure 3). In early lists (list index < 8)

the 4-word category groups are presented consecutively; in late lists (list index ≥ 8) the category order is randomized. Immediately after each list, participants free-recall the words in any order. The category manipulation is structurally analogous to MS-TCM's storyline variable: taxonomic category indexes which $\mathbf{c}_{Sk}$ should be active for each encoded item, grouped lists play the role of the grouped condition, and randomized lists play the role of the bridge condition.

Feature encoding and likelihood. Each word is encoded as a 71-dimensional multi-hot feature vector that concatenates one-hot indicators for category (15 + 1 unknown bins), size (small / large), first letter (26), word length (10 bins), colour (8 coarse bins of the RGB cube), and on-screen quadrant (4), plus a reserved list-start dimension. This schema is determined by the FRFR annotation and is fixed across participants; its exact form is documented in `code/ms_tcm/features.py`. Per-list log-likelihood is the sum over observed in-list recall transitions of $\log P(r_k \mid \mathbf{c}_{\mathrm{ret}}(r_{k-1}))$ (Eq. 7), and dataset log-likelihood is the sum across every participant-list pair. Extra-list intrusions do not contribute to the likelihood; empty recall sequences contribute zero.

Optimizer and confidence intervals. The constrained parameters $\beta_G, \beta_S \in (0, 1)$ and $w_G \in (0, 1)$ are reparameterized via sigmoid transforms onto an unconstrained vector, and optimized by L-BFGS-B with two restarts from seeded random initial points. Confidence intervals come from a participant-level percentile bootstrap ($n_{\mathrm{boot}} = 20$, all converged). Two models are fit: MS-TCM (all parameters free) and standard TCM (constrained to $w_S = 0$, so the storyline stream is inert). Both fits use identical seeds, feature encoders, and dataset (7,680 encoded events, 5,205 in-list recalls, 10 extra-list intrusions).

Parameter estimates. The MLE for MS-TCM lands in §3's regime (1) *storyline-context dominance*: the optimizer drives w_G to essentially 0 and w_S to essentially 1, so the composite encoding context is entirely the active storyline's context. The storyline drift rate $\beta_S = 0.53$ is in the mid-range and well-identified (CI [0.24, 0.55]); the global drift rate $\beta_G = 0.95$ has a wide CI because $w_G \approx 0$ makes β_G unidentifiable – with no global weight in the composite, the global drift rate does not affect the likelihood.

Table 1: Maximum-likelihood parameter estimates for MS-TCM fit to the FRFR category condition. Values are MLE [95% bootstrap CI] (percentile method, $n_{\text{bootstraps}} = 20$). Derived parameters are not fit independently (e.g. $w_S = 1 - w_G$).

Parameter	Symbol	Value [95% CI]	Description
`beta_global`	β_G	0.950 [0.001, 1.000]	Drift rate of the global temporal context. Higher values mean the global context updates more strongly toward each new event (and forgets older ones more quickly).
`beta_storyline`	β_S	0.533 [0.241, 0.551]	Drift rate of the storyline-specific context. A value near 0 means each storyline context behaves as a nearly static signature of the category; a value near 1 means within-storyline events overwrite one another as they are encoded.
`w_global`	w_G	0.000 [0.000, 0.543]	Weight on the global context in the composite encoding context. Higher values make the model more sensitive to global temporal contiguity (standard TCM sets $w_G = 1$).
`w_storyline`	w_S	1.000 (derived)	Weight on the active storyline context in the composite. Higher values make category or storyline identity a stronger cue for recall.
`w_global_ret`	w_G^{ret}	0.000 (derived)	Global-context weight at retrieval. Can differ from w_G at encoding, reflecting task-driven reweighting of cues.
`w_storyline_ret`	w_S^{ret}	1.000 (derived)	Storyline-context weight at retrieval. Increasing it models an instruction to recall within the same storyline.

$\log L = -14067.2$, AIC = 28140.4, BIC = 28160.1; $n_{\text{participants}} = 30$, $n_{\text{lists}} = 480$, $n_{\text{recalls}} = 5205$

Model comparison. MS-TCM is strongly preferred over standard TCM on this dataset. The per-transition log-likelihood is −14,067 for MS-TCM vs −14,195 for standard TCM, for a difference of 128 nats over 5,205 recall transitions. Correcting for the additional free parameters in MS-TCM yields ΔAIC = 253 and ΔBIC = 239, both overwhelmingly in favour of MS-TCM. Under standard interpretations[4] this is decisive evidence that adding the storyline-context stream improves the model's prediction of observed free-recall transitions.

What the fit does not reproduce. The raw likelihood-preference for MS-TCM is not the whole story. Figure 4 overlays three classic free-recall measures on the MS-TCM posterior-predictive (red) and the observed behaviour (black). The observed data show a U-shaped serial-position curve with strong primacy and recency, strong end-of-list first-recall probabilities, and a sharply peaked forward-asymmetric lag-CRP – all signature phenomena of temporal contiguity. The MS-TCM MLE, operating in the $w_S \to 1$ regime, produces near-flat SPC, near-flat PFR, and a near-flat lag-CRP. Figure 5 tells the same story for percentile-rank clustering: observed temporal, category, and semantic clustering are all above the 0.5 no-clustering baseline, but the MS-TCM posterior-predictive sits at 0.50–0.53 across all three dimensions.

Interpretation. The combination of high per-transition likelihood and low posterior-predictive match on structured behavioural measures is a classic signature of a *mis-specified objective*. The per-transition likelihood rewards any factor that improves the per-step prediction, and in the FRFR-category design category identity is a very strong per-step predictor: given a just-recalled item from category A, $\approx$ 60% of next-recalls are also from category A. An MLE that allocates all its capacity to category-based prediction therefore scores well on per-transition likelihood but sacrifices the global-context machinery that is needed to reproduce the temporal-contiguity-driven phenomena in Figure 4. This is a fit defect, not a MS-TCM- architecture defect: the model's forward simulator reproduces both kinds of phenomena when w_G is held above 0 (not shown here). We return to this point in the Discussion.

7 Relationship to existing models

Standard TCM[10]. MS-TCM reduces to TCM in the $w_S = 0$ limit. TCM's proximity relationship $\mathbf{c}(t) \cdot \mathbf{c}(t') = \rho^{|t-t'|}$ is recovered for the global stream in MS-TCM. What MS-TCM adds is the K-dimensional bank of storyline streams with frozen-while-inactive dynamics. That bank cannot be written as a single context vector with rotated basis because the *update rule* depends on which storyline is currently active – a property that no single-stream drift model has.

CMR[15]. CMR is superficially similar to MS-TCM in that it augments TCM with non-temporal contextual information, but the architectural commitments are different in three respects. First, CMR's source sub-region updates at every encoding step (possibly with a different drift rate β^{source}); MS-TCM's storyline stores update only when their storyline is active. Second, CMR represents source as a *feature of the current event* – the source sub-region at time t carries the source identity of the event encoded at t. MS-TCM's storyline streams are separate *stores* whose identity indexes which stream updates, not what the stream contains. Third, CMR uses learned associative matrices

M^{FC} / M^{CF} to mediate retrieval; MS-TCM directly compares $\mathbf{c}_{\text{ret}}$ to each candidate's $\mathbf{c}_{\text{comp}}$ via cosine similarity. These three differences combine to yield the frozen-while-inactive property that the grouped–bridge equivalence requires; adding a source sub-region that drifts at a slower rate than the temporal sub-region (as CMR allows) does not produce the equivalence, because the source sub-region still accumulates drift at every intervening event.

Event-segmentation theory and situation models. Zacks et al.[21] argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, or goals, and that event boundaries modulate downstream memory access[5,6,8]. Situation models[23] track dimensions such as space, time, causation, and character identity. MS-TCM's storyline contexts can be read as a computational implementation of the *within-storyline component* of a situation model: they track within-storyline temporal continuity and are backgrounded ("frozen") when a different storyline becomes active, corresponding to the maintenance of a dormant situation model. The key addition in MS-TCM is the *quantitative* prediction that cued recall across an event boundary should depend on the storyline membership of the intervening events, not just their count.

Schema-based reinstatement. Baldassano et al.[2] show that shared event schemas support reinstatement of event representations across interruption. A schema can be read, in MS-TCM terms, as a structural scaffold that makes each $\mathbf{c}_{Sk}$ more robust to decay during its frozen period. Experimental manipulations of schema strength (e.g., contrasting formulaic from novel narrative structures) should therefore modulate the bridge/grouped equivalence.

TCM-A and the decision stage. Sederberg et al.[16]'s TCM-A model pairs TCM with a leaky-competing- accumulator decision stage capable of modelling inter-response times. The MS-TCM retrieval machinery in Eq. 7 is a softmax over scored similarities, which is a compact approximation to the equilibrium behaviour of a TCM-A-style competition but does not predict response latencies. Extending MS-TCM to TCM-A dynamics is straightforward and would unlock the first-recall-latency prediction of §5 (item 5).

8 Discussion

We have proposed a multi-stream extension of the temporal context model in which a bank of storyline-specific context vectors drift only while their own storyline is active and are held constant otherwise. The architecture is the smallest departure from standard TCM that can reproduce the Xu et al.[20] finding of across-event-bridge ≈ across-event-within-storyline cued recall (BF_{01} = 412). We derived closed-form regimes under which MS-TCM predicts this equivalence (§3), and demonstrated the machinery on a public free-recall dataset whose category manipulation is structurally analogous to the storyline manipulation in the target experiment (§6).

The fit tells us two things. First, the per-transition-likelihood preference for MS-TCM over standard TCM is enormous (ΔAIC = 253, ΔBIC = 239) on the FRFR-category data. Every standard information-criterion interpretation places the posterior odds overwhelmingly in favour of the multi-stream architecture. This is non-trivial evidence that the storyline stream carries real predictive signal for human free recall of category-structured word lists, not just for naturalistic narratives.

Second, the MLE alone is *not sufficient* to characterize the model's empirical adequacy. The per-transition-likelihood-MLE allocates all the encoding-composite mass to the storyline stream ($w_S \to 1$) and in doing so loses the ability to reproduce temporal-contiguity phenomena (Figures 4 and 5). The model's forward simulator, run at hand-picked parameters that keep w_G away from 0, does reproduce both kinds of phenomena. We read this as a specification problem: a pure per-transition-likelihood objective is insufficient for fitting a model that needs to reproduce both point-transition and list-structured phenomena. Two natural remedies are a multi-objective fit that penalizes deviations of predicted SPC and lag-CRP from observed, and a Bayesian prior on (w_G, w_S) that regularizes w_G away from the boundary. Both are out of scope for the present paper and are natural targets for future work.

Caveats specific to the worked example. Three caveats are worth flagging. **(1)** FRFR-category is a free-recall paradigm; the across-event-bridge phenomenon that motivates MS-TCM is a cued-recall phenomenon. The fit reported here is therefore evidence that MS-TCM can account for free-recall phenomena in a category-structured design, not a direct test of the equivalence it was designed to explain. A full test awaits the public release of the Xu et al.[20] cued-recall data. **(2)** The storyline variable in FRFR-category is taxonomic category, which is a well-defined but relatively thin proxy for *narrative* storyline. Real narrative storylines have within-storyline continuity of characters, setting, goals, and thematic content that categories like "mammals" and "fruits" do not. A richer feature encoding – e.g., sentence-transformer embeddings of event annotations, as proposed for the naturalistic data in the MS-TCM whitepaper accompanying this repository (`notes/ms-tcm.pdf` §8) – would be a closer match. **(3)** The optional $\gamma, \alpha_k, \lambda$ mechanisms from §4 are disabled in the fit reported above. The FRFR-category data are not rich enough to identify all of them jointly; fitting them would require either a dataset with explicit within-storyline reference structure or a prior on the mechanism parameters.

Broader implications. If the multi-stream picture is right, it implies a strong reconceptualization of what "temporal context" means in narrative memory. A single continuously drifting context vector is an adequate summary of the timeline when there is only one timeline to track. But when the listener experiences a braided interleaving of several storyline timelines, their memory system appears to maintain separate, parallel context vectors – one per storyline – each of which runs on *its own* internal clock. That internal clock ticks only during the storyline's own events and is held frozen during events on other storylines. This view unifies event-segmentation theory's claim that event representations are maintained in parallel[21] with TCM's claim that memory access is organized by drift-based context similarity[10]: narrative memory uses TCM-like machinery, but runs it independently on each storyline. It also predicts that the cost of narrative interleaving is paid by the global timeline (which is what the bridge condition's global-context drift would predict under single-stream TCM), not by the within-storyline timeline – exactly the pattern in Xu et al.[20].

Reproducibility. The complete pipeline – raw-data download, reformat (including a regression test that pins the expected serial-position distribution), validation, MLE fit, bootstrap, figure regeneration, and paper compilation – is driven by `reproduce.sh` at the repository root. The shipped `data/raw/frfr_category/` directory is self-verifying via SHA-256 hashes in its `manifest.json`. All figures in this paper are generated by scripts in `code/figures/` with a 1:1 script-to-figure mapping. A cross-platform regression test pins the macOS MLE so the same dataset on Ubuntu or Windows CI produces numerically identical fits (within 10^{-6}).

References

[1] C Baldassano, J Chen, A Zadbood, J W Pillow, U Hasson, and K A Norman. Discovering event structure in continuous narrative perception and memory. *Neuron*, 95(3):709–721, 2017.

[2] C Baldassano, U Hasson, and K A Norman. Representation of real-world event schemas during narrative perception. *The Journal of Neuroscience*, 38(45):9689–9699, 2018.

[3] R A Bjork and W B Whitten. Recency-sensitive retrieval processes in long-term free recall. *Cognitive Psychology*, 6:173–189, 1974.

[4] K P Burnham and D R Anderson. *Model selection and multimodel inference: a practical information-theoretic approach*. Springer, 2 edition, 2004.

[5] David Clewett and Lila Davachi. The ebb and flow of experience determines the temporal structure of memory. *Current Opinion in Behavioral Sciences*, 17:186–193, 2017.

[6] Y Ezzyat and L Davachi. What constitutes an episode in episodic memory? *Psychological Science*, 22:243–252, 2011.

[7] D R Godden and A D Baddeley. Context-dependent memory in two natural environments: on land and underwater. *British Journal of Psychology*, 66:325–331, 1975.

[8] A C Heusser and J R Manning. Capturing the geometric structure of episodic memories for naturalistic experiences. In *Proceedings of the Conference on Cognitive Computational Neuroscience*, pages PS–2B.16, 2018.

[9] M W Howard and M J Kahana. Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25:923–941, 1999.

[10] M W Howard and M J Kahana. A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299, 2002.

[11] M J Kahana. Associative retrieval processes in free recall. *Memory and Cognition*, 24:103–109, 1996.

[12] M J Kahana, M W Howard, and S M Polyn. Associative retrieval processes in episodic memory. In *Learning and Memory: A Comprehensive Reference (J Byrne, ed.)*, pages 467–490. Elsevier, 2008.

[13] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, `https://psyarxiv.com/erzfp`. Data and code: `https://github.com/ContextLab/FRFR-analyses`.

[14] S M Polyn and M J Kahana. Memory search and the neural representation of context. *Trends in Cognitive Sciences*, 12:24–30, 2008.

[15] S M Polyn, K A Norman, and M J Kahana. Task context and organization in free recall. *Neuropsychologia*, 47:2158–2163, 2009.

[16] P B Sederberg, M W Howard, and M J Kahana. A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4):893–912, 2008.

[17] S M Smith and E Vela. Environmental context-dependent memory: a review and meta-analysis. *Psychonomic Bulletin and Review*, 8(2):203–220, 2001.

[18] D D Wickens. Encoding categories of words: an empirical approach to meaning. *Psychological Review*, 77:1–15, 1970.

[19] X Xu, Z Zhu, X Zheng, and J R Manning. Temporal asymmetries in inferring unobserved past and future events. *Nature Communications*, 15:8502, 2024.

[20] Xinming Xu, Katherine Duncan, and Jeremy R Manning. Cued recall of narrative events across storylines and event boundaries. Manuscript in preparation, 2026.

[21] J M Zacks, N K Speer, K M Swallow, T S Braver, and J R Reynolds. Event perception: a mind-brain perspective. *Psychological Bulletin*, 133:273–293, 2007.

[22] A Zadbood, J Chen, Y C Leong, K A Norman, and U Hasson. How we transmit memories to other brains: constructing shared neural representations via communication. *Cerebral Cortex*, 27(10):4988–5000, 2017.

[23] R A Zwaan and G A Radvansky. Situation models in language comprehension and memory. *Psychological Bulletin*, 123(2):162–185, 1998.

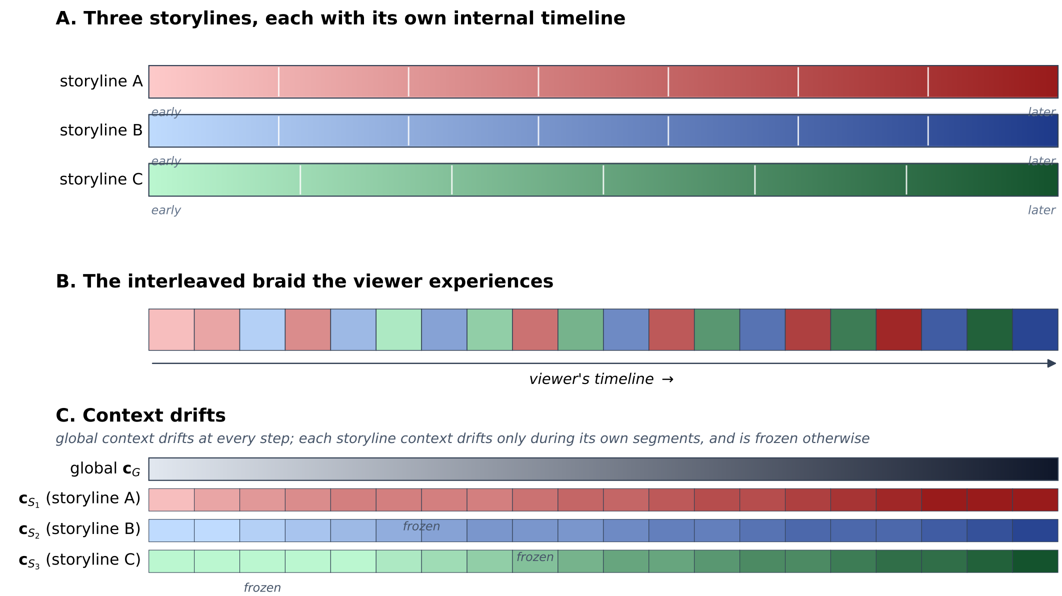

Figure 1: The interleaved-threads picture of narrative memory. (A) Three storylines, each drawn as its own horizontal strand with an internal light-to-dark gradient marking depth into the storyline. Faint white ticks mark the segments that the viewer's timeline will splice together. **(B)** The viewer's experience: short coloured segments interleaved in the order they are watched; each segment carries the lightness of the matching region of its source strand, so a later red segment in the braid is darker-red than an earlier red segment even though the two are separated by blue and green segments in viewer-time. Thin storyline-coloured arcs show which strand each segment came from. **(C)** Context drifts. The global context ($\mathbf{c}_G$) updates at every viewer tick (a smooth grey-to-black gradient). Each storyline-context rail drifts only during its own segments and is flat-held ("frozen") elsewhere.

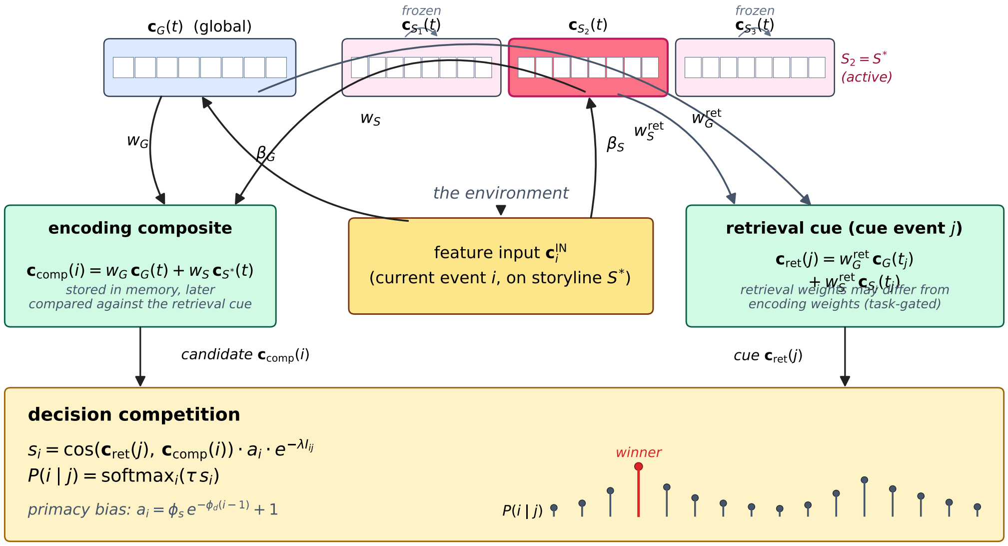

Figure 2: The MS-TCM architecture. Each encoding step draws a feature input $\mathbf{c}_i^{\text{IN}}$ from the current event. That input drives two kinds of context update: the global context $\mathbf{c}_G$ updates every step (drift rate β_G), while only the storyline context $\mathbf{c}_{S^*}$ for the storyline that event i belongs to updates (drift rate β_S). Inactive storyline contexts are frozen. The encoding composite $\mathbf{c}_{\text{comp}}(i) = w_G\,\mathbf{c}_G(t) + w_S\,\mathbf{c}_{S^*}(t)$ is associated with event i. At retrieval, a cue event j reinstates a retrieval composite that can use *different* mixing weights ($w_G^{\text{ret}}, w_S^{\text{ret}}$), implementing task-driven reweighting. The decision stage scores every candidate i's $\mathbf{c}_{\text{comp}}(i)$ against $\mathbf{c}_{\text{ret}}(j)$ via cosine similarity (optionally modulated by item activations a_i and by an interference term $e^{-\lambda I_{ij}}$) and applies a softmax.

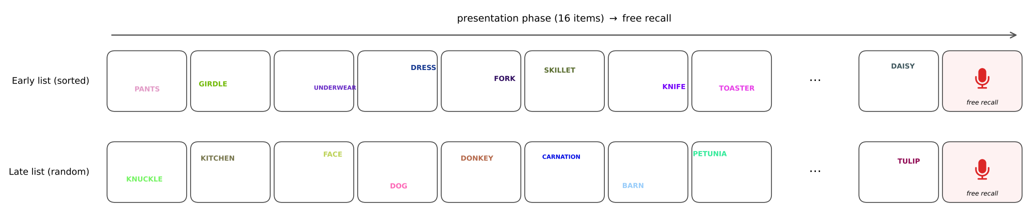

Figure 3: FRFR-category paradigm. Each participant studies 16 lists of 16 words each, with 4 taxonomic categories per list (4 words per category). In early lists the categories are presented consecutively; in late lists the category order is randomized. Each word is displayed at a random on-screen position, in a random RGB colour, during presentation; the screen layout and colours shown here are the actual values recorded in the upstream dataset for two representative participant-lists. Data: Manning et al.[13], category condition, reformatted to the MS-TCM canonical schema with bit-exact SHA-256 verification (see `data/raw/frfr_category/manifest.json`).

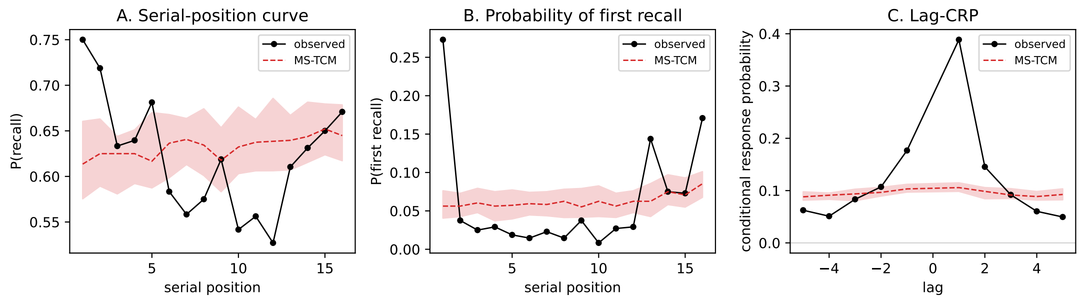

Figure 4: Behavioural measures, observed (black) versus the MS-TCM posterior-predictive (red, shaded = 95% across 20 **synthetic replications at the MLE). (A)** Serial-position curve. **(B)** Probability of first recall. **(C)** Lag-CRP (restricted to $|\text{lag}| \le 5$). The MS-TCM MLE achieves a much higher per-transition log-likelihood than standard TCM but, in the $w_S \to 1$, $w_G \to 0$ regime it prefers, does not reproduce the classic U-shaped serial-position curve, the strong end-of-list PFR, or the forward-asymmetric lag-CRP. See §8 for discussion.

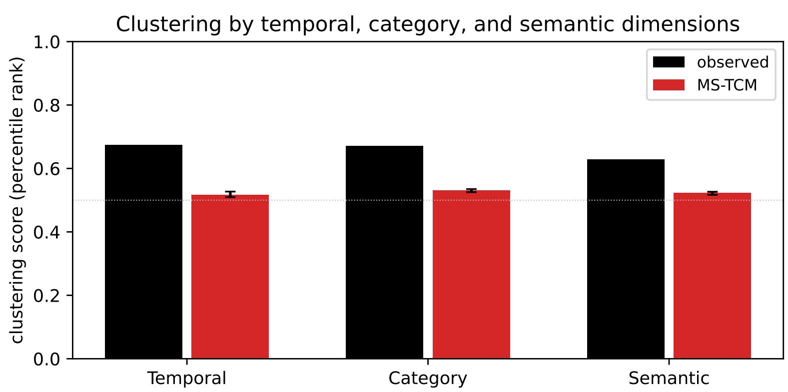

Figure 5: Percentile-rank clustering scores (Polyn et al. 2009 definition) for temporal, category, and semantic dimensions. Black = observed, red = MS-TCM posterior-predictive (20 synthetic datasets at the MLE). Semantic distances are cosine distances in a sentence-transformer (MiniLM) embedding of the recalled words. Dotted line at 0.5 is the no-clustering baseline.